Summary

Cross-Persona Summary

PERSONA MAP

Persona	Type	Frequency	Depth	Primary Need
Priya Venkataraman	Primary	Daily — high volume	Operational / deep	Explainable, auditable, overridable AI recommendations
Damien Okafor	Secondary	Daily — lower volume	Operational / specialist	Complete case handoff context + pattern-based search
Lorraine Thibodeau	Tertiary	Weekly / exception	Strategic / governance	Audit-ready compliance documentation & aggregate reporting
Marcus Sundaram	Admin / Operator	Periodic / reactive	Configuration & monitoring	Performance feedback loop + versioned configuration management

TENSION MAP

1. **Tension 1: Priya (efficiency) vs. Lorraine (caution)**

   Priya wants the auto-apply confidence threshold as low as operationally defensible to reduce her daily queue burden. Lorraine wants it high to limit regulatory exposure from automated decisions. Design implication: Marcus's performance dashboard must make this trade-off visible and evidence-based for both stakeholders — so threshold decisions are data-driven, not opinion-driven.

2. **Tension 2: Speed of resolution (Priya, Damien) vs. quality of customer contact (Dorothy)**

   Operational incentives push toward fast case closure. Dorothy's experience requires that cases needing outreach receive deliberate, prepared, single-contact communication — which takes longer. Design implication: the system must distinguish between cases that can close without customer contact and those where contact quality matters more than closure speed.

3. **Tension 3: Transparency for analysts (Priya) vs. auditability for governance (Lorraine)**

   Priya needs per-case decision reasoning visible in the transaction view. Lorraine needs aggregate decision distribution visible in a governance view. These are different audiences, different data granularities, and different interfaces. Design implication: build two distinct view tiers — analyst-level case view and director-level compliance dashboard — rather than a single interface trying to serve both.

4. **Tension 4: Investigator pattern knowledge (Damien) vs. system configuration (Marcus)**

   Damien accumulates institutional knowledge about edge case patterns through his investigations. Marcus needs that knowledge to calibrate configuration. But the pathway from Damien's observations to Marcus's configuration proposals to Lorraine's approval is currently informal and undocumented. Design implication: build a structured mechanism — a pattern library or structured case tagging system — that lets Damien's observations become Marcus's evidence.

PRIORITY GUIDANCE

The system must be designed primarily for Priya — the daily, high-volume analyst whose satisfaction determines whether the AI PoC delivers operational value. Her need for explainable, auditable, overridable recommendations is non-negotiable and shapes the core case-resolution UX. Damien's needs are secondary in design priority but critical in system architecture: the case handoff packet, persistent annotation, and decision trace features must be built to enable his work, even though they are not the primary UI surface. Lorraine's compliance requirements are non-negotiable constraints that apply globally across the system — they are not lower priority, they simply shape different layers of the product (audit logging, configuration governance, role-based access) rather than the transaction-level interface. Marcus's administrative needs are the enabling infrastructure: without them, the system cannot improve over time, but they do not compete with analyst-tier UX in the design sprint prioritisation. Dorothy's experience is shaped entirely by how well Priya and Damien are supported in making outbound contact deliberate and coordinated — designing for her means designing the communication layer, not the core resolution workflow.

PRIMARY PERSONA · Priya Venkataraman

PRIMARY PERSONA · Priya Venkataraman

> *"I need to clear my queue by the end of day — and I need to trust that the system isn't about to create a problem I'll have to clean up tomorrow."*

1. BIO

Priya Venkataraman, 34, is a Payment Operations Analyst at a mid-sized regional insurance carrier in Columbus, Ohio. She transitioned into this role four years ago from a bank reconciliation position at a credit union, and now manages 60–100 payment cases per day in a hybrid work setting. A significant share of her caseload lands in the miscellaneous or suspense bucket because senders omit policy numbers, use partial identifiers, or pay the wrong amount. Priya is usually the first person to hear a policyholder complaint about a payment that has sat unallocated for days, a position she finds professionally uncomfortable and operationally avoidable.

2. ROLE & CONTEXT

Field	Detail
Job / life role	Resolves unidentified and miscellaneous payments; applies, holds, or escalates each case; flags duplicates
Organisation / setting	Mid-sized insurance carrier; payment operations team of 8–12 analysts
Tech fluency	Medium — fluent in legacy insurance admin systems and CRMs; cautious about new AI tools until she trusts their outputs
Device context	Dual-screen desktop at office; laptop when remote
Access context	Continuous daily use, 8–9 hours; processes queue from open to close of business
Decision-making power	Decides on lower-risk cases; escalates high-risk or ambiguous payments; cannot override verified fraud flags

3. GOALS

1. **Clear the suspense queue accurately before close of business**
   Every unidentified payment must reach a disposition — Apply, Hold, or Escalate — by end of day. Errors that bounce back cost her double the time and create downstream service failures.
2. **Understand the reasoning behind every AI recommendation**
   Before acting on a system suggestion, Priya needs to see the underlying signals: name match score, amount variance, risk flag status, and policy match confidence. Without that, the recommendation is noise she must override.
3. **Offload clear-cut cases so she can focus on genuinely hard ones**
   Priya wants the AI to handle high-confidence, risk-free cases autonomously so she can invest full attention in the ambiguous or flagged cases that actually require human judgment.

4. PAIN POINTS & FRUSTRATIONS

1. **Opaque system decisions force 15–30 minutes of manual reconstruction per case**
   When the current system places a payment in suspense with no explanation, Priya must cross-check the policy database, re-run name comparisons, and review payment history herself. She has built a personal spreadsheet to track patterns the system never surfaces.
2. **No system-generated audit trail creates accountability risk**
   When a payment is disputed, she reconstructs her decision process from memory and personal notes. She fears a contested case will be escalated back to her without documentation she can defend.
3. **Queue spikes make prioritisation impossible**
   After billing cycle cut-offs, her queue can surge unpredictably. Without complexity-based triage, simple and complex cases are interleaved and she loses focus switching between them.
4. **Trust in AI outputs is entirely conditional on seeing the reasoning**
   If the system produces a recommendation she cannot verify, Priya overrides it manually — negating the efficiency gain the tool is supposed to provide.

5. BEHAVIOURS & MENTAL MODELS

1. **Triangulates across multiple signals before committing**
   Priya never acts on a single data point. Before applying or holding a payment she cross-checks sender name, expected amount, payment history, and policy status. She treats the AI as one of several sources, not as an authority.
2. **Front-loads easy cases on high-complexity days**
   On heavy days she consciously batches high-confidence cases first to clear cognitive bandwidth before tackling genuinely ambiguous ones — a sequencing strategy the current system does not support.

6. NEEDS & EXPECTATIONS

1. **Explainable recommendations with all signal values co-located - Show 5 Highest Impact Scores**
   Functional — every AI recommendation must display name match score, amount variance, risk flag detail, and policy match confidence in the same view as the recommended action.
2. **System-generated, immutable audit log per case - Backend only**
   Functional — the log must capture every input value, signal computed, recommendation made, and human action taken, timestamped, without requiring Priya to write notes.
3. **Complexity-based queue triage**
   Functional — the queue must be sortable by confidence score, scenario type, and risk level so Priya can allocate attention proportionally.

4. **Low-friction override and annotation controls**
   Functional / Emotional — overriding a recommendation must be as quick as accepting one, and annotations must be easy to add so the record reflects her actual reasoning.

8. DESIGN IMPLICATIONS

1. **Surface all reasoning signals in the case view — never just the label**
   Priya will not act on "APPLY" or "HOLD" alone. Every recommendation must include name score, amount variance, risk flags, and policy match confidence rendered as readable values in the same view.
2. **Build complexity-based triage into the queue UI**
   Sort and filter controls by confidence score, scenario type, and risk level are essential — not nice-to-have. Without them, Priya's cognitive load management defaults to a personal workaround.
3. **Auto-generate audit trail as the case progresses**
   The audit log must be system-written in real time, not reconstructed.
4. **Make override and annotate as frictionless as accept**
   Override controls must be primary-action affordances. If overriding requires more clicks than accepting, Priya will accept recommendations she isn't sure about rather than slow down her queue.

SECONDARY PERSONA · Damien Okafor

SECONDARY PERSONA · Damien Okafor

> *"Give me the full picture when you hand me a case — I shouldn't have to rebuild the investigation from scratch when someone else has already done half the work."*

1. BIO

Damien Okafor, 41, is a Senior Payment Investigations Specialist at the same carrier, based out of the Atlanta processing office. Twelve years in insurance operations — the last six focused on fraud investigation, misdirected payments, and escalated anomalies — have given him an encyclopaedic mental catalogue of edge-case patterns. He handles the cases that no one else can close: Scenario 4 no-match escalations, confirmed or suspected fraud flags, third-party payment disputes, and duplicate reversals requiring direct contact with policyholders or banks. Damien is respected team-wide for his pattern recognition, but consistently frustrated that escalated cases arrive without the investigation context already gathered at the analyst level.

2. ROLE & CONTEXT

Field	Detail
Job / life role	Investigates escalated, unresolved, and high-risk payments; contacts policyholders and third parties; provides final case determinations
Organisation / setting	Insurance carrier; specialist investigations sub-team within payment operations; liaisons with fraud and compliance functions
Tech fluency	High — experienced across multiple insurance admin platforms, fraud detection tools, and case management systems; open to AI assistance if evidence quality is demonstrable
Device context	Desktop workstation; phone for outbound contact with policyholders and financial institutions
Access context	Manages a personal case queue; receives escalations routed from Priya's tier; fewer cases, deeper per-case engagement
Decision-making power	High — final authority on escalated cases; authorised to contact external parties; can recommend policy-level actions including fraud review

3. GOALS

1. **Receive cases with full investigation context already attached**
   Cases should arrive with all computed signals, match scores, analyst action history, and any supporting notes so Damien can start where the analyst left off — not reopen a blank file.
2. **Identify systemic patterns across escalated cases to reduce future volume**
   Beyond resolving individual cases, Damien wants to surface recurring failure patterns — payers who habitually mismatch, third-party payment setups that confuse the system

3. **Maintain a defensible, time-stamped investigative record for every case**

   Because his decisions may face fraud review or regulatory scrutiny, every data point consulted, action taken, and rationale applied must be captured in an immutable system record — not personal notes.

4. PAIN POINTS & FRUSTRATIONS

1. **Cases arrive without the reasoning that triggered the escalation**

   Damien regularly receives cases with minimal context — a sender name, an amount, and a "could not match" status. He re-runs lookups Priya already performed, sometimes reaching different conclusions because her working notes don't travel with the case.

3. **Institutional knowledge about edge case patterns is entirely in his head**

   Damien has deep pattern memory about recurring case types, but none of it is captured in the system. When a similar case appears weeks later, another investigator — or Damien after a holiday — starts from zero.

4. **Fuzzy match suggestions are presented without algorithmic provenance**

   When the system surfaces a "best match at 72% similarity" for an unidentified payment (Scenario 4), Damien needs to know which fields were compared. Without that detail, he treats the suggestion as decorative, not diagnostic.

5. BEHAVIOURS & MENTAL MODELS

1. **Constructs a hypothesis before making any external contact**

   Before calling a policyholder or bank, Damien assembles a working theory from all available signals. He treats AI-generated correlation data as input to his hypothesis, not as the hypothesis itself.

2. **Documents investigation steps in real time, not in retrospect**

   Having had cases re-opened months after closure, Damien annotates his reasoning as he works. He expects tools to support in-progress annotation, not just final disposition recording.

3. **Uses personal pattern memory to shortcut ambiguous cases**

   When Damien sees a specific combination of signals — a business-name sender with an invoice reference paying into a personal policy — he immediately recognises the pattern from prior investigations. Currently this exists only in memory, not in any system-level pattern library.

4. **Risk-ranks his queue before touching a single case**

   Damien does not process in order of arrival. He scans the full queue, assigns risk priority, and addresses potential fraud cases immediately. The system currently presents a flat queue with no risk-level indication.

6. NEEDS & EXPECTATIONS

1. **A structured case handoff packet that includes all upstream context**

   Functional — escalated cases must arrive with a summary of all computed signals, name match details, amount variance, risk flags, analyst action history, and any annotations, rendered as a readable at-a-glance packet on case entry.

2. **Persistent investigator-level annotation on the case record**

Functional — Damien's investigation notes must become part of the permanent case record, attached to the case rather than stored in a separate system.

3. **Cross-case pattern search across historical escalations**
   Functional — a searchable pattern library, or at minimum cross-case keyword search, would let Damien surface relevant precedents without relying entirely on personal recall.
4. **Risk-level classification visible at the queue level**
   Functional — Damien's risk-ranking behaviour is manual because the queue presents no risk signal. Exposing confidence score or scenario type at the queue tier would eliminate this workaround.
5. **Algorithmic detail accompanying every fuzzy match candidate**
   Functional — any similarity score presented to Damien must include: which fields were compared, which algorithm was applied, and what specific values generated the score.

7. RELATIONSHIP TO OTHER PERSONAS

Damien is directly downstream of Priya — he receives every case she cannot resolve. He is accountable to Lorraine (Tertiary) for fraud-risk outcomes and investigation of SLA adherence. His findings from Scenario 4 and 5 cases represent the highest-stakes decisions in the system, and the quality of Priya's case handoff directly determines how efficiently he can resolve them. Marcus (Admin) depends on Damien's operational observations to identify configuration failures.

8. DESIGN IMPLICATIONS

1. **Similar to persona 1 but exclusive to escalated cases.**

TERTIARY PERSONA · Lorraine Thibodeau

TERTIARY PERSONA · Lorraine Thibodeau

> *"Every decision this system makes must be explainable to an auditor. If we can't prove what the machine did and why, we've traded one compliance problem for a bigger one."*

1. BIO

Lorraine Thibodeau, 52, is the Director of Payment Operations and Compliance at the carrier's Hartford, Connecticut headquarters. She has held operational leadership roles in insurance for over 20 years, including direct responsibility for the organisation's last major payment platform overhaul in 2017. Lorraine is methodical and risk-averse by professional formation: misapplied payments, undetected fraud, and unresolved suspense balances all represent regulatory exposure, and she manages the organisation's relationship with that exposure daily. She reads industry and regulatory news before her first meeting each morning and applies a compliance officer's frame to every technology evaluation. She is the executive sponsor of the AI PoC but does not use the system at the transaction level — her touchpoints are configuration approvals, escalation reports, and audit documentation.

2. ROLE & CONTEXT

Field	Detail
Job / life role	Oversees payment operations strategy and compliance; approves configuration thresholds; reviews audit reports and SLAs; signs off on AI system deployment
Organisation / setting	Insurance carrier headquarters; interfaces with legal, compliance, and external regulators
Tech fluency	Medium — understands system capabilities and limitations conceptually; relies on dashboards and reports rather than operational tooling
Device context	Laptop and mobile; primarily report and dashboard consumption
Access context	Weekly or exception-based; reviews escalation summaries, SLA metrics, and audit logs rather than individual cases
Decision-making power	Very high — approves all configuration thresholds, authorises deployment, and is accountable to regulators and senior leadership for payment operations outcomes

3. GOALS

1. **Ensure every AI-assisted decision is fully auditable**
   Every recommendation, human approval, and override must be captured in a complete, time-stamped, immutable audit trail that would satisfy an insurance regulator or internal auditor on demand.

2. **Set and govern the right automation / oversight balance**
   Lorraine's goal is not to eliminate human judgment — it is to eliminate human effort on decisions that are demonstrably low-risk, while preserving full accountability on every case the AI touches.
3. **Demonstrate regulatory compliance to external auditors**
   She must be able to produce documentation showing that the AI's decision logic is defined, bounded, and explainable to the same standard expected of a human decision-maker under insurance payment regulations.

5. BEHAVIOURS & MENTAL MODELS

1. **Evaluates new technology through a risk-and-failure-mode lens first**
   Before asking "what efficiency does this create?", Lorraine asks "what fails, who is accountable, and what is the audit trail when it does?" She will not approve deployment until these questions have documented answers.
2. **Reads aggregate metrics as system health diagnostics**
   Lorraine does not review individual cases. She reads escalation rates, hold rates, override frequencies, and SLA adherence as leading indicators of whether the system is configured correctly.
3. **Requires formal change control for any threshold modification**
   In Lorraine's mental model, changing a confidence threshold from 90% to 85% is a regulated configuration change that must be documented, reviewed, and approved — not an informal operational adjustment.

6. NEEDS & EXPECTATIONS

1. **Aggregate audit dashboard showing AI decision distribution over time**
   Functional — Lorraine needs a report view (not case view) showing decision counts by outcome (Apply/Hold/Escalate), confidence score distributions, override rates, and human approval rates, with date-range filtering.
2. **Version-controlled configuration log with change rationale fields**
   Functional — every threshold change must be captured with who made the change, when, why, and what the expected operational impact was. This log must be export-ready for regulatory review.
3. **Role-based access separating her oversight view from analyst workflow**
   Functional — Lorraine's interface must surface compliance and governance information without mixing her into the transaction-level workflow designed for Priya.

7. RELATIONSHIP TO OTHER PERSONAS

Lorraine does not interact with the system at the case level. She receives aggregate performance reports that reflect Priya's queue outcomes and Damien's investigation results. She is the authority who defines the configuration parameters both personas operate within — setting the 90% auto-apply threshold, the 72-hour duplicate window, and the amount of variance tolerances. Her approval gates all threshold changes Marcus proposes.

8. DESIGN IMPLICATIONS

1. **Build a compliance-grade audit log as a first-class system output, not an afterthought**
   The AI PoC brief already lists "complete audit trail" as an explicit system requirement. This must be per-decision, system-generated, immutable, and export-ready — not reconstructed from logs.
2. **Version-control all configuration with rationale and approver fields**
   Configuration parameters must be treated like code: versioned, with change history, rationale, and the identity of who approved each change stored alongside each version.
3. **Create a separate aggregate dashboard tier for Lorraine's role**
   Decision distribution trends, override rates, and SLA compliance must be available in a dedicated view that does not require Lorraine to navigate the analyst-level case queue.
4. **Document the AI decision logic in audit-presentable format**
   The rule sets behind Scenario 1–5 routing must be rendered as human-readable documentation that can be presented to a regulator as evidence that the system operates within defined, accountable parameters.
5. **Enforce role-based access control across the system**
   Lorraine's access level must reflect her oversight role — she should not be able to (or need to) intervene in individual case decisions, and her view must not be cluttered with transaction-level operations detail.

ADMIN / OPERATOR PERSONA ·

Marcus Sundaram

ADMIN / OPERATOR PERSONA · Marcus Sundaram

> *"I need to tune the system based on what's actually happening in the queue — not what we modelled when we designed it. The thresholds are hypotheses, not facts."*

1. BIO

Marcus Sundaram, 38, is a Business Systems Analyst and Operations Technology Lead at the carrier's Philadelphia regional office. He was the primary internal project lead for the AI PoC, coordinating between the vendor team, IT infrastructure, and payment operations. Marcus designed the configuration threshold system documented in Scenarios 1–5 — the name-match percentage bands, amount variance tolerances, and duplicate detection window — and he maintains them. He approaches his role with an engineer's emphasis on measurable outcomes and a data analyst's instinct for leading indicators. Marcus cycles to work most days and thinks of system performance monitoring the same way: incremental, consistent, and evidence-based. He is not a daily transaction-level user, but he is accountable when the system behaves unexpectedly.

2. ROLE & CONTEXT

Field	Detail
Job / life role	Configures and maintains AI decision thresholds; monitors system performance; diagnoses anomalies; bridges operations team and technology
Organisation / setting	Insurance carrier IT / operations technology function; collaborates with payment ops, compliance, and vendor teams
Tech fluency	High — comfortable with data pipelines, rule configuration, SQL reporting, and system log analysis
Device context	Desktop and laptop; accesses configuration dashboards, monitoring tools, and raw system logs
Access context	Periodic access for performance review cycles; reactive access when Priya or Damien flags an anomaly; involved in every threshold change process
Decision-making power	Proposes threshold changes and configuration adjustments; implements once Lorraine approves; no unilateral production configuration authority

3. GOALS

1. **Keep decision quality aligned with real operational patterns as they evolve**
   The threshold values in Scenarios 1–5 were calibrated on modelled assumptions. Marcus monitors live performance to detect drift and proposes evidence-based adjustments before output quality degrades visibly.
2. **Diagnose unexpected system behaviours without digging through raw logs**
   When Priya or Damien reports a recommendation that seems wrong,

3. **Create a feedback loop so operational outcomes improve configuration over time**

   Every time Priya overrides a recommendation, that override is a data point about where the system's logic diverges from human judgment. Marcus wants that signal captured and aggregatable — not discarded after case closure.

4. PAIN POINTS & FRUSTRATIONS

1. **Override data from analysts is invisible to system performance monitoring**

   When Priya overrides an AI recommendation, the override is recorded as a case disposition but not as a signal the system can aggregate. Marcus cannot determine how often, in which scenario types, or at which confidence bands humans are diverging from AI outputs.

3. **Anomaly diagnosis requires interpretive work on raw system logs**

   System logs are structured for operational reliability monitoring, not for human-readable decision audit. Diagnosing why a specific payment was routed to Scenario 3 instead of Scenario 1 requires significant manual log interpretation.

5. BEHAVIOURS & MENTAL MODELS

1. **Monitors leading indicators — escalation rate, override frequency — before they surface as complaints**

   A sudden spike in escalation volume or a drop in auto-apply rate signals miscalibration before it becomes a visible problem. Marcus reads these as early-warning instruments and prefers to act before Priya or Lorraine notices an issue.

2. **Tests threshold changes against historical case data before promoting to production**

   Before any configuration change goes live, Marcus runs a back-test against a recent slice of cases to estimate the projected impact on decision distribution. He will not make a live change without this estimate.

3. **Maintains a personal threshold change log because the system has no memory of its own configuration history**

   Marcus's personal log — date, threshold changed, old value, new value, motivating evidence, observed outcome — is a critical institutional record that lives in a personal spreadsheet rather than in the system's configuration layer.

6. NEEDS & EXPECTATIONS

1. **Version-controlled configuration management with rationale fields**

   Functional — every threshold parameter must have a version history, and each version must capture: who changed it, when, from what value to what value, and why. This log must be system-maintained, not manually kept.

2. **Aggregate performance dashboard segmented by scenario type**

   Functional — Marcus needs to see per-scenario case volume, decision distribution (Apply/Hold/Escalate), override frequency, and confidence score spread, with time-range filtering and trend lines.

3. **Override data structured and queryable, not discarded at case closure**

   Functional — analyst and investigator overrides must be captured as structured events (scenario type, original recommendation, override action, annotated reason if provided) that Marcus can query and aggregate.

4. **Human-readable decision trace for anomaly diagnosis**
   Functional — for any individual case, Marcus must be able to view: what input values were received, which scenario path was selected, which branch conditions were evaluated, and what the final recommendation logic produced. Plain language, not raw logs.
5. **Staging or simulation mode for testing configuration changes**
   Functional — the ability to run a proposed threshold change against a set of historical cases and see the projected impact on decision distribution before promoting to production.

7. RELATIONSHIP TO OTHER PERSONAS

Marcus is the technical steward of the system that Priya and Damien use daily. He receives anomaly reports from both personas and translates operational observations into configuration proposals for Lorraine's approval. He is the conduit through which Damien's pattern-recognition knowledge and Priya's override behaviour can flow back into system improvement — but only if the right data capture infrastructure is built.

8. DESIGN IMPLICATIONS

1. **Treat configuration as a versioned, system-maintained record — not a manual spreadsheet**
   The configuration layer must have native version control with change history, rationale fields, and approver identity. Removing this from manual spreadsheets to a system-maintained record is foundational.
2. **Build an admin-level performance dashboard with per-scenario breakdown**
   Marcus's monitoring behaviour is sophisticated but currently relies on aggregate signals with no granularity. Per-scenario performance views — with override rates, confidence distributions, and trend lines — make his leading-indicator monitoring precise.
3. **Capture analyst overrides as structured, queryable performance events**
   This is the most critical feedback loop the system is missing. Override events must include: the original recommendation, the override action, the scenario type, and the confidence score at time of override. This data makes threshold calibration evidence-based.
4. **Provide a human-readable decision trace alongside raw logs**
   For each case, a plain-language decision path trace (input values → scenario routing → branch evaluation → recommendation) must be available to Marcus for anomaly diagnosis without requiring log file interpretation skills.
5. **Build a staging environment for threshold change testing**
   The ability to back-test a proposed configuration change against historical cases before production deployment transforms threshold changes from high-risk events into evidence-driven, low-risk adjustments.